

◦ Aim:

- To find the wavelength of He-Ne laser light using a millimetre scale as a grating.

◦ Apparatus:

- He-Ne laser, Vernier calliper, metre scale, graph paper, etc.

◦ Theory:

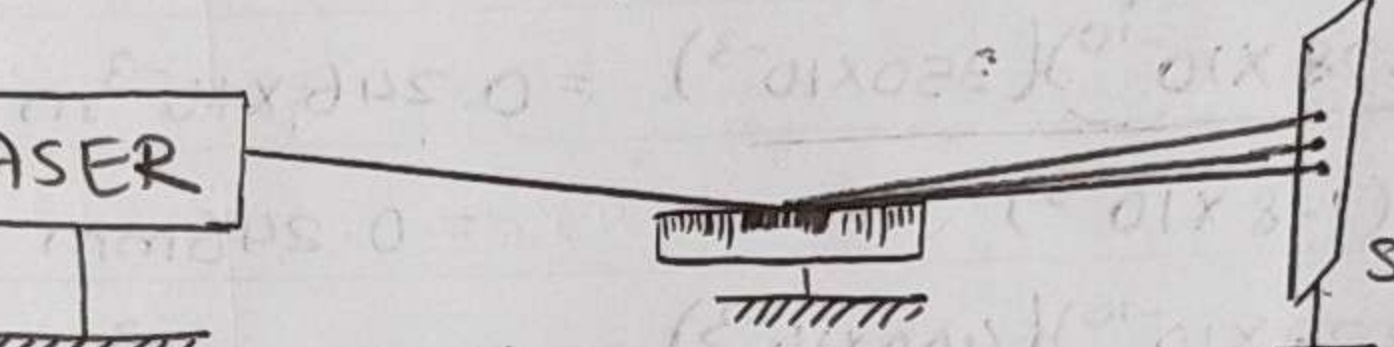
- The bending of light about sharp edges of an opaque object or a narrow slit is called as 'diffraction'.
- Schawlow in 1965 performed this experiment using a vernier calliper and He-Ne laser to find the wavelength of laser light by studying the diffraction pattern obtained from millimetre scale of a ruler.
- The unexpanded laser beam is allowed to fall at the grazing angle ($i = 87^\circ$) on the vernier calliper placed on a horizontal table and the diffraction pattern is observed at a distance of 3-4 metres from scale.
- The diffraction taking place on engraving of scale follows:

$$d(\sin i - \sin \theta_m) = m\lambda \quad \text{--- (1)}$$

where d = grating constant
 i = angle of incidence
 θ_m = angle corresponding to order 'm'.
 λ = wavelength of light.

10
15 100 180

min $z = 0$ [Figure (1)] (50 x 1.1)



01.05 [Experimental Setup]

- observations:

- $d = 0.04 \text{ cm}, z_0 = 75 \text{ cm}$

Sr. No.	Distance of diffraction spots	$A_m = y_m^2$	$A_m - A_0 = y_m^2 - y_0^2$	$\lambda = \frac{d}{m} \left[\frac{y_m^2 - y_0^2}{2z_0^2} \right]$
1	$y_0 = 5.5 \text{ cm}$	30.25	0	0
2	$y_1 = 7.2 \text{ cm}$	51.84	21.59	$6.48 \times 10^{-5} \text{ cm}$
3	$y_2 = 9.2 \text{ cm}$	86.64	54.39	$8.42 \times 10^{-5} \text{ cm}$
4	$y_3 = 9.8 \text{ cm}$	96.04	65.79	$6.60 \times 10^{-5} \text{ cm}$
5	$y_4 = 10.8 \text{ cm}$	116.64	86.39	$6.48 \times 10^{-5} \text{ cm}$
6	$y_5 = 11.8 \text{ cm}$	139.24	108.99	$6.55 \times 10^{-5} \text{ cm}$
7	$y_6 = 12.7 \text{ cm}$	161.29	131.04	$6.57 \times 10^{-5} \text{ cm}$
8	$y_7 = 13.6 \text{ cm}$	184.96	151.71	$6.66 \times 10^{-5} \text{ cm}$
9	$y_8 = 14.3 \text{ cm}$	204.49	174.24	$6.55 \times 10^{-5} \text{ cm}$
10	$y_9 = 15 \text{ cm}$	225	194.75	$6.43 \times 10^{-5} \text{ cm}$

o calculations:

- we know that, $\lambda = \frac{d}{m} \left[\frac{y_m^2 - y_0^2}{2z_0^2} \right] = \frac{0.04}{m} \left[\frac{y_m^2 - y_0^2}{2(75)^2} \right] \text{ cm.}$

(i) $\lambda_1 = \frac{0.04}{1} \left[\frac{21.59}{2 \times 5625} \right] = 6.48 \times 10^{-5} \text{ cm.}$

(ii) $\lambda_2 = \frac{0.04}{2} \left[\frac{54.39}{2 \times 5625} \right] = 8.42 \times 10^{-5} \text{ cm}$

$$\lambda_3 = \frac{0.04}{3} \left[\frac{65.79}{2 \times 5625} \right] = 6.60 \times 10^{-5} \text{ cm}$$

$$\lambda_4 = \frac{0.04}{4} \left[\frac{86.39}{2 \times 5625} \right] = 6.48 \times 10^{-5} \text{ cm}$$

$$\lambda_5 = \frac{0.04}{5} \left[\frac{108.99}{2 \times 5625} \right] = 6.55 \times 10^{-5} \text{ cm}$$

$$\lambda_6 = \frac{0.04}{6} \left[\frac{131.04}{2 \times 5625} \right] = 6.57 \times 10^{-5} \text{ cm}$$

$$\lambda_7 = \frac{0.04}{7} \left[\frac{154.71}{2 \times 5625} \right] = 6.66 \times 10^{-5} \text{ cm}$$

$$\lambda_8 = \frac{0.04}{8} \left[\frac{174.24}{2 \times 5625} \right] = 6.55 \times 10^{-5} \text{ cm}$$

$$\lambda_9 = \frac{0.04}{9} \left[\frac{194.75}{2 \times 5625} \right] = 6.43 \times 10^{-5} \text{ cm}$$

- $\therefore \lambda_{\text{avg}} = \frac{\lambda_1 + \lambda_2 + \dots + \lambda_9}{9} = \frac{6.48 + 8.42 + 6.60 + 6.48 + 6.55 + 6.57 + 6.66 + 6.55 + 6.43}{9} \times 10^{-5} \text{ cm}$

$$\Rightarrow \lambda_{\text{avg}} = 6.748 \times 10^{-5} \text{ cm} = 6748 \text{ \AA}$$

- $\therefore \% \text{ error} = \left| \frac{\lambda_{\text{avg}} - \lambda_{\text{th}}}{\lambda_{\text{th}}} \right| \times 100 = \left| \frac{6748 - 6328}{6328} \right| \times 100 = 6.63\%$

- From figure (1), $i + \alpha \pm \pi/2 \Rightarrow i = -\alpha + \pi/2$

$$\theta_m + \beta_m = \pi/2 \Rightarrow \theta_m = \pi/2 - \beta_m.$$

From eqⁿ (1), $d [\sin(\pi/2 - \alpha) + \sin(\pi/2 - \beta_m)] = m\lambda$

$$\therefore d [\cos \alpha + \cos \beta_m] = m\lambda \quad \text{--- (ii)}$$

- When $m=0 \Rightarrow \cos \alpha = \cos \beta_m \Rightarrow \alpha = \beta_0$. --- (iii).

- In $\triangle POM$, $\sin \beta_m = \frac{OM}{PM} = \frac{y_m}{\sqrt{y_m^2 + z_0^2}} = \frac{y_m}{z_0}$ --- ($z_0 \gg y_m$).

Now, as $\cos \beta_m = \sqrt{1 - \sin^2 \beta_m}$.

$$\therefore \cos \beta_m = \sqrt{1 - \frac{y_m^2}{z_0^2}} = \left(1 - \frac{y_m^2}{z_0^2}\right)^{1/2} = 1 - \frac{1}{2} \left(\frac{y_m^2}{z_0^2}\right) \Rightarrow \cos \beta_m = 1 - \frac{y_m^2}{2z_0^2} \quad \text{--- (iv)}$$

- From (ii) and (iv), $d [\cos \alpha - \cos \beta_m] = m\lambda \Rightarrow d [\cos \beta_0 - \cos \beta_m] = m\lambda$.

$$\therefore d \left[1 - \frac{y_0^2}{2z_0^2} - 1 + \frac{y_m^2}{2z_0^2}\right] = m\lambda \Rightarrow d \left[\frac{y_m^2 - y_0^2}{2z_0^2}\right] = m\lambda.$$

$$\text{Hence, we get } \lambda = \frac{d}{m} \left[\frac{y_m^2 - y_0^2}{2z_0^2}\right].$$

• Procedure:

- Fix the positions of laser, screen, and calculate the distance from Vernier scale to screen (z_0). Place graph paper on screen.
- Adjust the laser so that it incidents the vernier calliper at a grazing angle and forms a diffraction pattern on the screen.

- Remove the vernier calliper and mark the point of direct beam.
- Now, place the calliper back and mark all the diffraction spots. Mark the point midway to the direct beam and 1st point.
- Measure the rest of the points from this new points.
- Calculate the value of λ from each spot and calculate the average λ . Also, find the % error.
- Precautions:
 - No laser light should enter the eyes.
 - Laser exposure on skin must not be for too long.
- Applications:
 - He-Ne laser is used for barcode reading, eye treatment, etc.
- Result:
 - Wavelength of He-Ne laser using millimetre scale is $6748\text{\AA}$.
- Conclusion:
 - The wavelength of He-Ne laser is found out to be $6748\text{\AA}$ with an error of 6.63% from the actual value of $6328\text{\AA}$.